

CStats - Lab #1

February 24, 2026

Exercise 1.

Suppose that a school has 20 classes: 16 with 25 students in each, three with 100 students in each, and one with 300 students, for a total of 1000 students.

(a) What is the average class size?

(b) Select a student randomly out of the 1000 students. Let the random variable X equal the size of the class to which this student belongs, and define the probability mass function of X .

(c) Find $\mathbb{E}(X)$, the expected value of X . Does this answer surprise you ?

Solution 1.

(a) The average class size is

$$\frac{(16)(25) + (3)(100) + (1)(300)}{20} = 50;$$

(b)

$$f(x) = \begin{cases} 0.4, & x = 25, \\ 0.3, & x = 100, \\ 0.3, & x = 300. \end{cases}$$

(c)

$$\mathbb{E}(X) = 25(0.4) + 100(0.3) + 300(0.3) = 130.$$

Exercise 2.

A coin comes up heads with probability p and tails with probability $1 - p$. We do not know p . We flip the coin 100 times and observe X heads. Derive $\hat{p}_{ML}(X)$, the ML estimator for p .

Solution 2.

The unknown here is p .

We want to derive:

$$\hat{p}_{\text{ML}}(X = x) = \arg \max_p \mathbb{P}\{X = x \mid p\}.$$

X is a r.v., where $X \sim \text{Binomial}(100, p)$. Thus,

$$\mathbb{P}\{X = x \mid p\} = \binom{100}{x} p^x (1-p)^{100-x}, \text{ where } 0 \leq x \leq 100$$

To find the value of p which maximizes the above expression, we need to differentiate it:

$$\begin{aligned} 0 &= \frac{d}{dp} \mathbb{P}\{X = x \mid p\} \\ &= \frac{d}{dp} \binom{100}{x} p^x (1-p)^{100-x} \\ &= p^x \cdot (100-x) \cdot (1-p)^{99-x} \cdot (-1) + (1-p)^{100-x} \cdot x \cdot p^{x-1} \end{aligned}$$

This yields

$$\begin{aligned} p^x (100-x) (1-p)^{99-x} &= (1-p)^{100-x} \cdot x \cdot p^{x-1} \\ p(100-x) &= (1-p) \cdot x \\ p(100-x+x) &= x \\ p &= \frac{x}{100} \end{aligned}$$

It can be verified that $p = \frac{x}{100}$ indeed maximizes $\mathbb{P}\{X = x \mid p\}$. So

$$\hat{p}_{\text{ML}}(X = x) = \arg \max_p \mathbb{P}\{X = x \mid p\} = \frac{x}{100}$$

Since the above holds for any arbitrary value of $0 \leq x \leq 100$, it follows that

$$\hat{p}_{\text{ML}}(X) = \frac{X}{100}$$

Exercise 3.

Let $X_1, X_2, \dots, X_n$ be i.i.d. samples of r.v. X with unknown finite mean and variance. Let $\bar{X}$ denote the sample mean. Define

$$\bar{S}^2 \equiv \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Prove that $\bar{S}^2$ is **not** an unbiased estimator of $\text{Var}(X)$. Follow these steps:

(a) Prove that

$$\mathbb{E}[\bar{S}^2] = \frac{1}{n} \sum_{i=1}^n \text{Var}(X_i - \bar{X}).$$

(b) Show that

$$\text{Var}(X_i - \bar{X}) = \frac{n-1}{n} \text{Var}(X).$$

(c) Combine (a) and (b) to show that

$$\mathbb{E}[\bar{S}^2] = \frac{n-1}{n} \text{Var}(X).$$

Solution 3.

$$\begin{aligned} \mathbb{E}[\bar{S}^2] &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(X_i - \bar{X})^2] \\ &= \frac{1}{n} \sum_{i=1}^n \text{Var}((X_i - \bar{X})) + \mathbb{E}[X_i - \bar{X}]^2 \\ &= \frac{1}{n} \sum_{i=1}^n \text{Var}((X_i - \bar{X})) + (\mathbb{E}[X] - \mathbb{E}[\bar{X}])^2 \\ &= \frac{1}{n} \sum_{i=1}^n \text{Var}((X_i - \bar{X})) \end{aligned}$$

(b)

$$\begin{aligned} \text{Var}(X_i - \bar{X}) &= \text{Var}\left(X_i - \frac{X_1 + X_2 + \cdots + X_n}{n}\right) \\ &= \text{Var}\left(\frac{nX_i - (X_1 + X_2 + \cdots + X_n)}{n}\right) \\ &= \text{Var}\left(\frac{X_1 + X_2 + \cdots + X_n - nX_i}{n}\right) \\ &= \frac{1}{n^2} \cdot \text{Var}(X_1 + X_2 + \cdots + (1-n)X_i + \cdots + X_n) \\ &= \frac{1}{n^2} ((n-1) \text{Var}(X) + (1-n)^2 \text{Var}(X)) \\ &= \frac{\text{Var}(X)}{n^2} \cdot (n-1 + 1 - 2n + n^2) \\ &= \frac{\text{Var}(X)}{n^2} \cdot n(n-1) \\ &= \frac{n-1}{n} \text{Var}(X) \end{aligned}$$

(c)

$$\begin{aligned}\mathbf{E}[S^2] &= \frac{1}{n} \sum_{i=1}^n \text{Var}((X_i - \bar{X})) \\ &= \frac{1}{n} \sum_{i=1}^n \frac{n-1}{n} \text{Var}(X) \quad \text{by (b)} \\ &= \frac{1}{n} \cdot n \cdot \frac{n-1}{n} \cdot \text{Var}(X) \\ &= \frac{n-1}{n} \cdot \text{Var}(X)\end{aligned}$$

Exercise 4.

Team A has probability p of beating team B. We do not know p , but we can see that in the last 10 games played between A and B, team A won seven games and team B won three games. Assume that every game has a unique winner and that games are independent. Based on this information, formulate and compute the ML estimator for p .

Solution 4.

We define $\hat{p}_{\text{ML}}(X)$ to be the MLE of p . Here X is the number of games won by team A out of 10.

What we want is

$$\hat{p}_{\text{ML}}(X = 7)$$

We want to derive:

$$\hat{p}_{\text{ML}}(X = x) = \arg \max_p \mathbb{P}\{X = x \mid p\}.$$

X is a r.v., where $X \sim \text{Binomial}(10, p)$. Thus

$$\mathbb{P}\{X = x \mid p\} = \binom{10}{x} p^x (1-p)^{10-x}, \text{ where } 0 \leq x \leq 10$$

To find the value of p which maximizes the above expression, we need to differentiate it:

$$\begin{aligned}0 &= \frac{d}{dp} \mathbb{P}\{X = x \mid p\} \\ &= \frac{d}{dp} \binom{10}{x} p^x (1-p)^{10-x} \\ &= p^x \cdot (10-x) \cdot (1-p)^{9-x} \cdot (-1) + (1-p)^{10-x} \cdot x \cdot p^{x-1}\end{aligned}$$

This yields

$$p^x(10-x)(1-p)^{99-x} = (1-p)^{10-x} \cdot x \cdot p^{x-1}$$

$$p(10-x) = (1-p) \cdot x$$

$$p(10-x+x) = x$$

$$p = \frac{x}{10}$$

It can be verified that $p = \frac{x}{10}$ indeed maximizes $\mathbb{P}\{X = x \mid p\}$. So

$$\hat{p}_{\text{ML}}(X = x) = \arg \max_p \mathbb{P}\{X = x \mid p\} = \frac{x}{10}.$$

Hence

$$\hat{p}_{\text{ML}}(X = 7) = \frac{7}{10}.$$

Exercise 5.

A casino suspects that one of its roulette tables may be malfunctioning. To investigate, the casino records the net gain (in CHF) per round for a random sample of $n = 64$ rounds played by different customers. The sample yields:

$$\bar{X} = -2.5 \text{ CHF}, \quad s = 4.0 \text{ CHF}.$$

- Using the Central Limit Theorem, explain why we can assume that $\bar{X}$ follows an approximately normal distribution, regardless of the underlying distribution of individual round outcomes.
- Construct a 95% confidence interval for the true mean net gain per round μ . (Use $z_{0.025} = 1.96$.)
- Under normal operating conditions, the expected net gain for the player is -4 CHF per round (i.e., the house edge). The casino wants to test whether the table is deviating from this expected value. Formulate a two-sided hypothesis test at significance level $\alpha = 0.05$ and state your conclusion.
- If the casino wants the 95% confidence interval to have a total width of at most 1 CHF (i.e., margin of error ≤ 0.5 CHF), what is the minimum number of rounds it must observe? Assume the same standard deviation $s = 4.0$.

Solution 5.

(a) By the Central Limit Theorem, if $X_1, X_2, \dots, X_n$ are i.i.d. random variables with finite mean μ and finite variance σ^2 , then for sufficiently large n :

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{d} N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{as } n \rightarrow \infty.$$

With $n = 64 \geq 30$, this approximation holds well regardless of the shape of the population distribution. This is powerful because individual round outcomes are highly non-normal — a player either wins a large payout or loses their wager — yet the sampling distribution

of $\bar{X}$ is still approximately normal.

(b) The 95% confidence interval for μ is:

$$\bar{X} \pm z_{0.025} \cdot \frac{s}{\sqrt{n}} = -2.5 \pm 1.96 \cdot \frac{4.0}{\sqrt{64}} = -2.5 \pm 1.96 \cdot 0.5 = -2.5 \pm 0.98.$$

$$\boxed{95\% \text{ CI} = [-3.48, -1.52]}$$

We are 95% confident that the true mean net gain per round falls between -3.48 and -1.52 CHF.

(c) We test:

$$H_0 : \mu = -4 \quad \text{vs} \quad H_1 : \mu \neq -4.$$

Since $n = 64$ is large, we use the Z-test. The test statistic is:

$$Z = \frac{\bar{X} - \mu_0}{s / \sqrt{n}} = \frac{-2.5 - (-4)}{4.0 / \sqrt{64}} = \frac{1.5}{0.5} = 3.0.$$

The critical values at $\alpha = 0.05$ (two-sided) are $\pm z_{0.025} = \pm 1.96$. Since $|Z| = 3.0 > 1.96$, we **reject** H_0 .

Equivalently, the p -value is:

$$p\text{-value} = 2 \cdot \mathbf{P}\{Z > 3.0\} \approx 2 \times 0.0013 = 0.0026 < 0.05.$$

There is statistically significant evidence that the true mean net gain per round is **not** -4 CHF. The players are losing less than expected, suggesting the table may indeed be malfunctioning (or that a player is cheating). Note that $\mu_0 = -4$ falls outside the 95% CI from part (b), which is consistent with rejection.

(d) We require the margin of error $E \leq 0.5$:

$$E = z_{0.025} \cdot \frac{s}{\sqrt{n}} \leq 0.5.$$

Solving for n :

$$\begin{aligned} \sqrt{n} &\geq \frac{z_{0.025} \cdot s}{E} = \frac{1.96 \times 4.0}{0.5} = 15.68, \\ n &\geq 15.68^2 = 245.86. \end{aligned}$$

Rounding up: $\boxed{n = 246}$.

This illustrates that the required sample size grows with the square of the desired precision — halving the margin of error requires roughly quadrupling the number of observed rounds.